**Experiment 8: Top 5 Best-Selling Products Using Pandas**

**Aim**

To identify the top five products sold in the highest quantities during the past month using Pandas DataFrame operations.

**Algorithm**

1. Import the Pandas library.
2. Create a DataFrame containing product names and quantities sold.
3. Group the data by product name.
4. Calculate the total quantity sold for each product using sum().
5. Sort the quantities in descending order using sort_values().
6. Select the top five products using head(5).
7. Display the results.

**Code**

```
import pandas as pd

# Sales data for the past month

sales_data = pd.DataFrame({

   "Product": [

      "Laptop", "Mouse", "Keyboard", "Mouse",

      "Monitor", "Laptop", "Headphones", "Keyboard",

      "Mouse", "Monitor"

   ],

   "Quantity_Sold": [10, 50, 20, 30, 15, 5, 25, 10, 40, 20]

})

# Calculate total quantity sold for each product

product_sales = sales_data.groupby(

   "Product"

)["Quantity_Sold"].sum()
```

```
# Find the top 5 best-selling products
top_5 = product_sales.sort_values(
    ascending=False
).head(5)
print("Top 5 best-selling products:")
print(top_5)
```

**Input**

| Product | Quantity Sold |
|---|---|
| Laptop | 10 |
| Mouse | 50 |
| Keyboard | 20 |
| Mouse | 30 |
| Monitor | 15 |
| Laptop | 5 |
| Headphones | 25 |
| Keyboard | 10 |
| Mouse | 40 |
| Monitor | 20 |

**Output**

```
Top 5 best-selling products:
Product
Mouse         120
Keyboard       30
Monitor        35
Headphones     25
Laptop         15
```

Note: The correct descending order for the sample data is shown below. The code will automatically display this order.

Top 5 best-selling products:

Product

Mouse 120

Monitor 35

Keyboard 30

Headphones 25

Laptop 15

```
====== RESTART: C:/Users/sivak/Desktop/FODS LAB experiments codes/exp 8.py =====
Top 5 best-selling products:
Product
Mouse         120
Monitor        35
Keyboard       30
Headphones     25
Laptop         15
Name: Quantity_Sold, dtype: int64
```

**Result**

Thus, the five best-selling products were identified by grouping products, summing their quantities sold, and sorting the results in descending order.